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अनुभाग 208 एक दीवार वाले बेकिंग-अवन

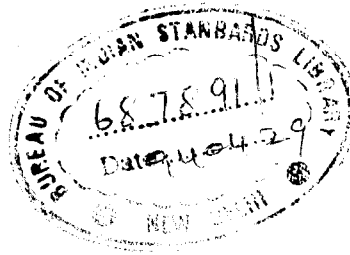
Indian Standard

SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES

PART 2 PARTICULAR REQUIREMENTS

Section 208 Single Walled Baking Ovens

UDC 64.06-83 : 641.535.3 : 614.895



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Price Group 4

FOREWORD

This Indian Standard was adopted by the Bureau of Indian Standards, after the draft finalized by the Electrical Appliances Sectional Committee had been approved by the Electrotechnical Division Council.

This Indian Standard covers the general and safety requirements of single walled baking ovens. This standard does not cover the performance requirements, which are covered under a separate standard IS 8985 : 1978 'Specification for single walled baking ovens'.

This standard is to be read in conjunction with IS 302-1 (1979) 'Safety of household and similar electrical appliances : Part 1 General requirements'. For the sake of convenience, the clauses of this standard correspond to those of IS 302-1 (1979), instead of reproducing full text of each clause. Clauses of IS 302-1 (1979) which are applicable (which means that relevant provisions of the clause apply) or not applicable and the sub-clauses or portions thereof which are not applicable are indicated as under:

- a) In case of a clause where it is applicable or not applicable, the wording used is 'This clause of IS 302-1 (1979) is applicable/not applicable'.
- b) In case of sub-clause or part thereof 'Not applicable'.

Wherever a sub-clause of IS 302-1 (1979) is to be replaced by a new text, it has been indicated as under:

'Replacement — followed by the next text.'

Any addition to the existing provisions of a sub-clause of IS 302-1 (1979) has been indicated as under:

'Addition — followed by the text of the additional matter.'

Clauses/tables which are additional to those of IS 302-1 (1978) are numbered starting from **101** and additional sub-clauses are numbered with the main clause number followed by **101, 102** etc, for example, **7.101**.

Should however, any deviation exist between IS 302-1 (1979) and this standard, the provisions of the latter shall apply.

While preparing this standard, assistance has been derived from the following publications issued by the International Electrotechnical Commission:

IEC Pub 335-1 (1976) Safety of household and similar electrical appliances : Part 1 General requirements

IEC Pub 335-2-6 (1986) Safety of household and similar electrical appliances : Part 2 Particular requirements for cooking ranges, cooking tables, ovens and similar appliances for household use.

The Indian Standard which are necessary adjuncts to this standard are given in Annex AA.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (revised)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Indian Standard

SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES

PART 2 PARTICULAR REQUIREMENTS

Section 208 Single Walled Baking Ovens

1 SCOPE

This clause of IS 302-1 (1979) is applicable except as follows:

1.1 Replacement

This standard covers general and safety requirements for Single Walled Baking Ovens for baking and grilling designed for household use, their rated voltage not exceeding 230 volts ac single phase or dc.

1.2 Replacement

This standard does not apply to:

- Double walled cooking ovens and other ovens which are covered by IS 302-2-6 (1993).

For appliances intended to be used in vehicles or on board ships or aircraft, additional requirements may be necessary.

2 TERMINOLOGY

2.1 This clause of IS 302-1 (1979) is applicable except as follows:

2.29 Replacement

Conditions of Adequate Heat Discharge — The load that apply when a heating appliance is operated under normal conditions of use. Baking ovens and other similar appliances for baking and grilling are operated so that the temperature at centre of heater surface is maintained at $250 \pm 15^{\circ}\text{C}$, either by means of a thermostat or by switching the supply 'on' and 'off'. If the thermostat does not allow temperature to reach 250°C , the appliance is operated with the thermostat at the highest setting.

Additional Sub-clauses

2.101 Baking Pan

A sheet or tray which can be withdrawn and on which food items, such as cake, biscuits can be baked.

2.102 Oven

A chamber which can be closed, and incorporates heating units intended for heating the chamber, in which baking and/or grilling is carried out.

2.103 Automatic Oven

An oven which is thermostatically controlled.

2.104 Non-automatic Oven

An oven which is not thermostatically controlled. It may however be provided with a thermal cut-out.

2.105 Temperature of Oven

The temperature anywhere in the central position of baking space, the space being the cube of 5 cm around the geometric centre of oven.

2.106 Baking Space

The space within an oven which is above the lowest position of baking oven and not less than 5 cm from heating surface.

2.107 Oven Grill

An oven which is provided with a grilling arrangement with a heating element fixed near the top of the baking space.

3 GENERAL REQUIREMENT

This clause of IS 302-1 (1979) is applicable except as follows:

3.1 Para 2 Not applicable.

4 GENERAL NOTES ON TESTS

This clause of IS 302-1 (1979) is applicable except as follows:

4.8, 4.10, 4.13 and 4.14 not applicable.

5 RATING

This clause of IS 302-1 (1979) is applicable except as follows:

5.2 Replacement

The rated input shall not exceed 2.0 kW.

6 CLASSIFICATION

This clause of IS 302-1 (1979) is applicable except as follows:

7 MARKING

This clause of IS 302-1 (1979) is applicable.

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7.101 Every oven shall be marked indelibly and clearly in accordance with 27.1 of IS 302-1 (1979) on its outer surface or on a nameplate firmly attached to it. A clear marking indicating the different switch positions and correspond rating in watts shall be provided where multi-top switches are provided. In addition to above a marking stating whether the oven is insulation 'OT' uninsulated type shall be there.

7.102 Every oven shall be supplied with the necessary instructions booklet including precautions to be taken for its proper use. The instruction shall describe in detail the manner of carrying out the various baking operations which are possible with the oven and shall in particular indicate the best position in the oven for different types of baking, type of vessel to be used, clearing instruction and the appropriate thermostat setting for various baking operation where a thermostat is provided.

7.103 The single walled baking ovens may also be marked with the Standard Mark.

7.103.1 The use of the Standard Mark is governed by the provisions of Bureau of Indian Standards Act, 1986 and the Rules and Regulations made thereunder. The details of conditions under which the licence for the use of Standard Mark may be granted to manufacturers or producers may be obtained from the Bureau of Indian Standards.

8 PROTECTION AGAINST ELECTRIC SHOCK

This clause of IS 302-1 (1979) is applicable.

8.2, 8.3 and 8.7 Not applicable.

9 STARTING OF MOTOR-OPERATING APPLIANCES

This clause of IS 302-1 (1979) is not applicable.

10 INPUT AND CURRENT

This clause of Part 1 is applicable.

10.2 Not applicable.

11 TEMPERATURE RISE

This clause of IS 302-1 (1979) is applicable.

11.5 and 11.6 Not applicable.

11.8 Replacement

The temperature rise of handle, knobs and grips and the like which, in normal use, are held for short periods and are made of moulded material shall not exceed 55°C.

11.9 and 11.10 Not applicable

12 OPERATION UNDER OVERLOAD CONDITIONS OF APPLIANCES WITH HEATING ELEMENTS

This clause of IS 302-1 (1979) is applicable except as follows:

12.3 Not applicable.

13 ELECTRICAL INSULATION AND LEAKAGE CURRENT AT OPERATING TEMPERATURE

This clause of IS 302-1 (1979) is applicable.

14 RADIO AND TELEVISION INTERFERENCE SUPPRESSION

This clause of IS 302-1 (1979) is applicable.

15 MOISTURE RESISTANCE

This clause of IS 302-1 (1979) is applicable except as follows:

15.1 Not applicable.

15.2 Addition

Ordinary and drip-proof appliances are also subjected to the splash test with the apparatus shown in Fig. 11 of IS 302-1 (1979).

15.4 Replacement

Instead of the periods specified, the following applies:

Appliances are placed in the humidity cabinet for a period of 48 h.

Addition

If it is not possible to place the whole appliance in the humidity cabinet, parts containing electrical components are tested separately, taking into account the conditions which occur in the appliance.

16 INSULATION RESISTANCE AND ELECTRIC SHOCK STRENGTH (AFTER HUMIDITY TREATMENT)

This clause of IS 302-1 (1979) is applicable except as follows:

16.3 Not applicable.

17 OVERLOAD PROTECTION

This clause of IS 302-1 (1979) is applicable.

18 ENDURANCE

(*Under Consideration*).

19 ABNORMAL OPERATION

This clause of IS 302-1 (1979) is applicable except as follows:

19.1 Replacement

Appliances shall be so designed that the risk of fire, mechanical damage impairing safety or the protection against electric shock as a result of abnormal or careless operation is obviated as far as is practicable.

Compliance is checked by the tests of 19.2 to 19.4.

If, in any of the tests, a non-self-resetting thermal cut-out operates, a heating element ruptures or if the current is otherwise interrupted before steady conditions are established, the heating period is considered to be ended, but if the interruption is due to the rupture of a heating element or of an intentionally weak part, the relevant test is repeated on a second sample which shall then also comply with the conditions specified in 19.11. Rupture of heating element or of an intentionally weak part in the second sample will not in itself entail a rejection.

Fuses, thermal cut-outs, overcurrent releases or the like incorporated into the appliance, may be used to provide the necessary protection.

If more than one of the tests are applicable for the same appliance, these tests are made consecutively.

Provisions of 19.11 apply to all appliances.

20 STABILITY AND MECHANICAL HAZARDS

This clause of IS 302-1 (1979) is applicable except as follows:

20.1 Addition

Covers, lids and accessories are placed in the most unfavourable positions.

20.2 Not applicable.

21 MECHANICAL STRENGTH

This clause of IS 302-1 (1979) is applicable.

21.2 to 21.4 Not applicable.

22 CONSTRUCTION

This clause of IS 302-1 (1979) is applicable except as follows:

22.1 Replacement

Appliances shall be of Class I construction.

Compliance is checked by inspection and by the relevant tests.

22.15, 22.18, 22.20, 22.24, 22.25, 22.29 and 22.32 Not applicable.

Additional Sub-clauses

22.101 Heating Chamber

The interior of the oven lining, sheet metal oven runners and baffles, drip-trays, and the inner panel of the oven door when made out of mild steel, shall be finished in vitreous enamel having resistance to thermal shock and heat such that it shall comply with the requirements laid down in IS 3972 : 1968. These may also be made of commercial purity aluminium (*see* IS 737 : 1986) in half hard condition duly anodized or stainless steel or any other suitable material or finish.

The heating chamber shall be so constructed and reinforced that baking pan is adequately supported and the material of heating chamber does not warp buckle when heated to maximum temperature of oven.

Interior surfaces of the oven should be such that they lend themselves for early and thorough cleaning even after continuous use.

Interior surfaces of the baking oven should be such that they lend themselves for early and thorough cleaning even after continuous use.

The ovens shall be provided with the suitable viewing arrangement.

22.102 Vents

One or more suitable vents shall be provided near top of oven, so arranged that they deflect fumes and moisture away.

22.103 Baking Pan and Tray

Every oven shall be supplied with at least one tray. The baking pan and tray shall be made of aluminium or any other suitable material taking into consideration that it does not warp on heating.

22.104 Heating Element

The heating element shall be so designed and located that the heating condition required for baking can be obtained and local over-heating or burning of the foodstuffs avoided. Use of mineral filled sheathed heating elements conforming to IS 4159 : 1983 is recommended. However, if bare heating coils supported in a tubing is used, steatite ceramic shall be used for supporting the heating coil.

22.105 Wiring and Components

The wiring and all the components used in an oven shall be constructed and fixed in such a manner that they shall be able to withstand continuously the temperature to which they are exposed when the oven is operated at its maximum temperature. All the wiring and components shall be suitable protected from spillage or overflow from vessels in the oven.

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22.106 Thermostats

Where a thermostat is used it shall comply with IS 4165 : 1991. An indicator shall be provided to give an indication when the contacts of thermostat have a voltage and current rating not less than the heating element it controls.

22.107 Handles

The oven shall be provided with handles of adequate size so that possibility of touching the body while handling the oven is avoided. The handles shall be made of heat insulating material for safe handling of the oven.

22.108 Switches

Manually operated switches if used in an oven shall comply with the relevant provisions of IS 3854 : 1988. Switches and controls shall be located or protected so that they are not subjected to mechanical injury, spillage from baking or collection of grease.

22.109 Heat Control

Oven heating elements of 750 watts and over shall be controlled by a switch having not less than three heat steps, by an energy regulator, or by a thermostat. A non-automatic oven shall be provided with a temperature indicator. In case of grill type of ovens provisions for separate switching arrangement of the grilling element.

23 INTERNAL WIRING

This clause of IS 302-1 (1979) is applicable.

24 COMPONENTS

This clause of IS 302-1 (1979) is applicable except as follows:

24.9 and **24.10** Not applicable.

24.11 Replacement

Thermal cut-outs shall provide all phase disconnection from the supply.

Compliance is checked by inspection.

25 SUPPLY CONNECTION AND EXTERNAL FLEXIBLE CABLES AND CORDS

This clause of IS 302-1 (1979) is applicable.

25.4 Addition

Type X attachment shall be used for appliances delivered without a cord.

25.6 Replacement

Instead of the types of power supply cord specified, the following applies:

Power supply cords shall be oil resistant sheathed

flexible cable not lighter than ordinary polychloroprene or other equivalent synthetic elastomer-sheathed cord [see IS 9968 (Part 1) : 1988].

25.6 Addition

The length of the supply cord measured between the entry point into the appliance and the entry point into the plug and cord guard included, shall be not less than 2.0 m if supplied with plug, or shall be not less than 2.05 m if supplied without plug.

25.10 Not applicable.

26 TERMINALS FOR EXTERNAL CONDUCTORS

This clause of IS 302-1 (1979) is applicable.

27 PROVISION FOR EARTHING

This clause of IS 302-1 (1979) is applicable except as follows:

27.2 Addition

Small fixed exposed metal parts, for example name plates and the like, are not required to be in electrical contact with the terminal.

28 SCREWS AND CONNECTIONS

This clause of IS 302-1 (1979) is applicable.

30 RESISTANCE TO HEAT, FIRE AND TRACKING

This clause of IS 302-1 (1979) is applicable.

31 RESISTANCE TO RUSTING

This clause of IS 302-1 (1979) is applicable.

32 RADIATION HAZARDS

This clause of IS 302-1 (1979) is applicable.

101 TESTS

101.1 Type Test

The tests specified in Table 101 shall constitute the type tests and shall be carried out on two samples of Single Walled Baking Ovens of the same type and rating selected preferably at random from a regular production lot. Before commencement of the test, the steam cookers shall be visually examined and inspected for obvious visual defects in respect of components, parts and their assembly, construction, mechanical hazards, markings, provision of suitable terminals for supply connections, earthing and the effectiveness of screws and connections. The external surface finish shall be even and free from finishing defects.

NOTE — See also 4.2.

Table 101 Schedule of Type Tests
(Clause 101.1)

Sl No.		Clause Reference
i)	Protection against electric shock	8
ii)	Input	10
iii)	Temperature rise	11
iv)	Operation under overload conditions	12
v)	Electrical insulation and leakage current at operating temperature	13
vi)	Radio and television interference suppression	14
vii)	Moisture resistance	15
viii)	Insulation resistance and electric strength (after humidity treatment)	16
ix)	Overload protection	17
x)	Endurance (Under consideration)	18
xi)	Abnormal operation	19
xii)	Stability and mechanical hazards	20
xiii)	Mechanical strength	21
xiv)	Construction	22
xv)	Internal wiring	23
xvi)	Components	24
xvii)	Cord grip and cord guard	25
xviii)	Terminals for external conductors	26
xix)	Provision for earthing	27
xx)	Screws and connections	28
xxi)	Creepage distance and clearance	29
xxii)	Resistance to heat time and tracking	30
xxiii)	Resistance to rusting	31
xxiv)	Radiation hazards	32

101.1.1 Criteria of Acceptance

Both samples shall successfully pass all the type tests for proving conformity with the requirements

of the standard. If any of the samples fails in any of the type tests, the testing authority at its discretion, may call for fresh samples not exceeding twice for original number and subject them again to all tests or to the test(s) in which failure(s) had occurred. No failure should be permitted in the repeat test(s).

101.2 Acceptance Tests

The following shall constitute the acceptance tests:

Test	Clause Reference
a) Protection against electric shock	8
b) Input	10
c) Temperature rise	11
d) Electrical insulation and leakage current operating temperature	13
e) Moisture resistance	15
f) Insulation resistance and electric strength (after humidity treatment)	16
g) Provision for earthing	27

NOTE — For the purpose of acceptance tests, the humidity treatment shall be done for 24 hours while conducting the test for moisture resistance (15).

101.2.1 A recommend sampling procedure for acceptance tests is given in Appendix B of IS 302-1 (1979).

101.3 Routine Test

The following shall constitute the routine tests:

Test	Clause Reference
a) Protection against electric shock	8
b) High voltage	13.3.2 of IS 302-1 (1979)
c) Provision for earthing	27

ANNEX A

TABLES OF TYPE TEST

Appendix A of IS 302-1 (1979) is not applicable.

ANNEX B

(Clause 101.2.1)

SAMPLING PROCEDURE FOR ACCEPTANCE TESTS

Appendix B of IS 302-1 (1979) is applicable.

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ANNEX C

ELECTRONIC CIRCUITS

Appendix C of IS 302-1 (1979) is applicable if electronic circuits are used.

ANNEX D

MEASUREMENT OF TEMPERATURE WITH THERMOMETERS

Appendix D of IS 302-1 (1979) is applicable.

ANNEX E

ALTERNATIVE TESTS FOR PROTECTED MOTOR UNITS

Appendix E of IS 302-1 (1979) is applicable.

ANNEX F

IMPACT TEST APPARATUS

Appendix F of IS 302-1 (1979) is applicable.

ANNEX G

THERMAL CONTROLS AND OVERLOAD RELEASES

Appendix G of IS 302-1 (1979) is applicable.

ANNEX H

MEASUREMENT OF CREEPAGE DISTANCES AND CLEARANCES

Appendix H of IS 302- (1979) is applicable.

ANNEX J

TEST FOR FIRE RESISTING PROPERTIES

Appendix J of IS 302-1 (1979) is applicable.

ANNEX K

**BNF JET TEST FOR DETERMINATION OF THICKNESS OF COPPER
AND NICKEL PLATING**

Appendix K of IS 302-1 (1979) is not applicable.

ANNEX L

APPROXIMATE MEASUREMENT OF THICKNESS OF CHROMIUM ON NICKEL, STEEL AND COPPER

Appendix L of IS 302-1 (1979) is not applicable.

ANNEX AA

(Foreword)

LIST OF REFERRED INDIAN STANDARDS

<i>IS No</i>	<i>Title</i>	<i>IS No.</i>	<i>Title</i>
302-1 (1979)	Safety of household and similar electrical appliances: Part 1 General requirement (<i>fifth revision</i>)	3972 : 1968	Method of test for enamelware, vitreous enamelware
		4159 : 1983	Mineral filled sheathed heating elements (<i>second revision</i>)
737 : 1986	Wrought aluminium and aluminium alloy sheet and strip for general engineering purposes (<i>third revision</i>)	4165 : 1991	Thermostats for general purpose electric ovens (<i>first revision</i>)
		9968	Elastomer insulated cables :
3854 : 1988	Switches for domestic and similar purposes (<i>first revision</i>)	(Part 1) : 1988	Part 1 For working voltages up to and including 1 100 Volts (<i>first revision</i>)

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Amendments are issued to standards as the need arises on the basis of comments. Standards are also reviewed periodically; a standard along with amendments is reaffirmed when such review indicates that no changes are needed; if the review indicates that changes are needed, it is taken up for revision. Users of Indian Standards should ascertain that they are in possession of the latest amendments or edition.

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Amendments Issued Since Publication

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